



# NOAA's Heliophysics Missions and Data

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NOAA National Centers for Environmental Information

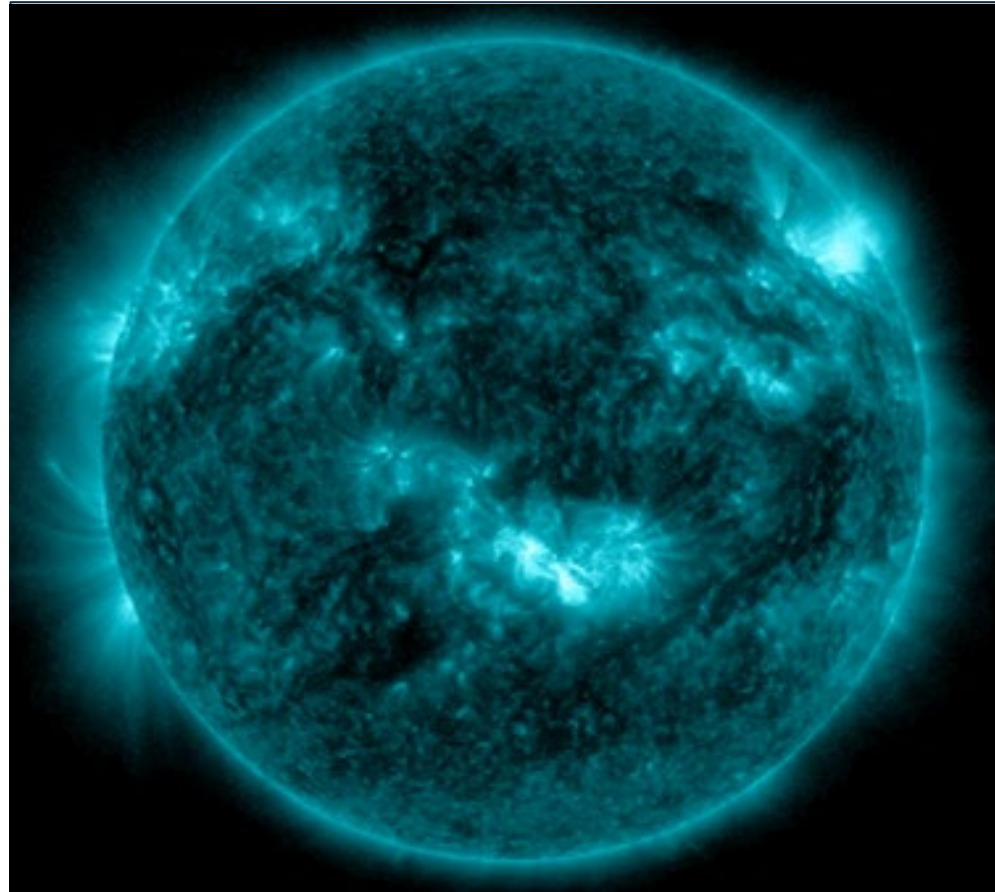
William Rowland, Janet Machol, Juan Rodriguez, Donald Schmit, Brian Kress, Jim Spann, Irfan Azeem, Yaireska Collado-Vega, Erin Lynch, Dimitrios Vassiliadis, Nai-Yu Wang

# NOAA's Space Weather Mission

*Safeguarding Society with Actionable Space Weather Information*



Weather, Water, and  
Climate Strategy  
FY 2023-2027



## NOAA provides:

- Observational Data
- Impact-based Decision Support
- Science and Model Development
- Data Management



# NOAA's Space Weather Groups

**NESDIS Office of  
Space Weather Observations  
(SWO) & Office of Satellite and  
Product Operations (OSPO)**



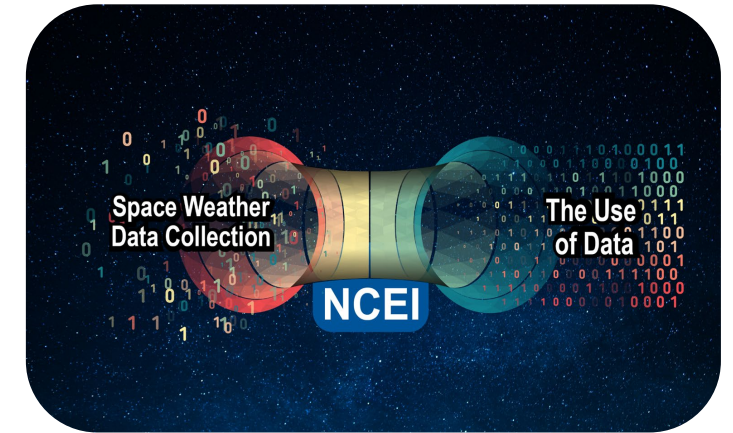
**Space-based Observations:  
System acquisition and  
satellite operations**

**NWS Space Weather  
Prediction Center  
(SWPC)**



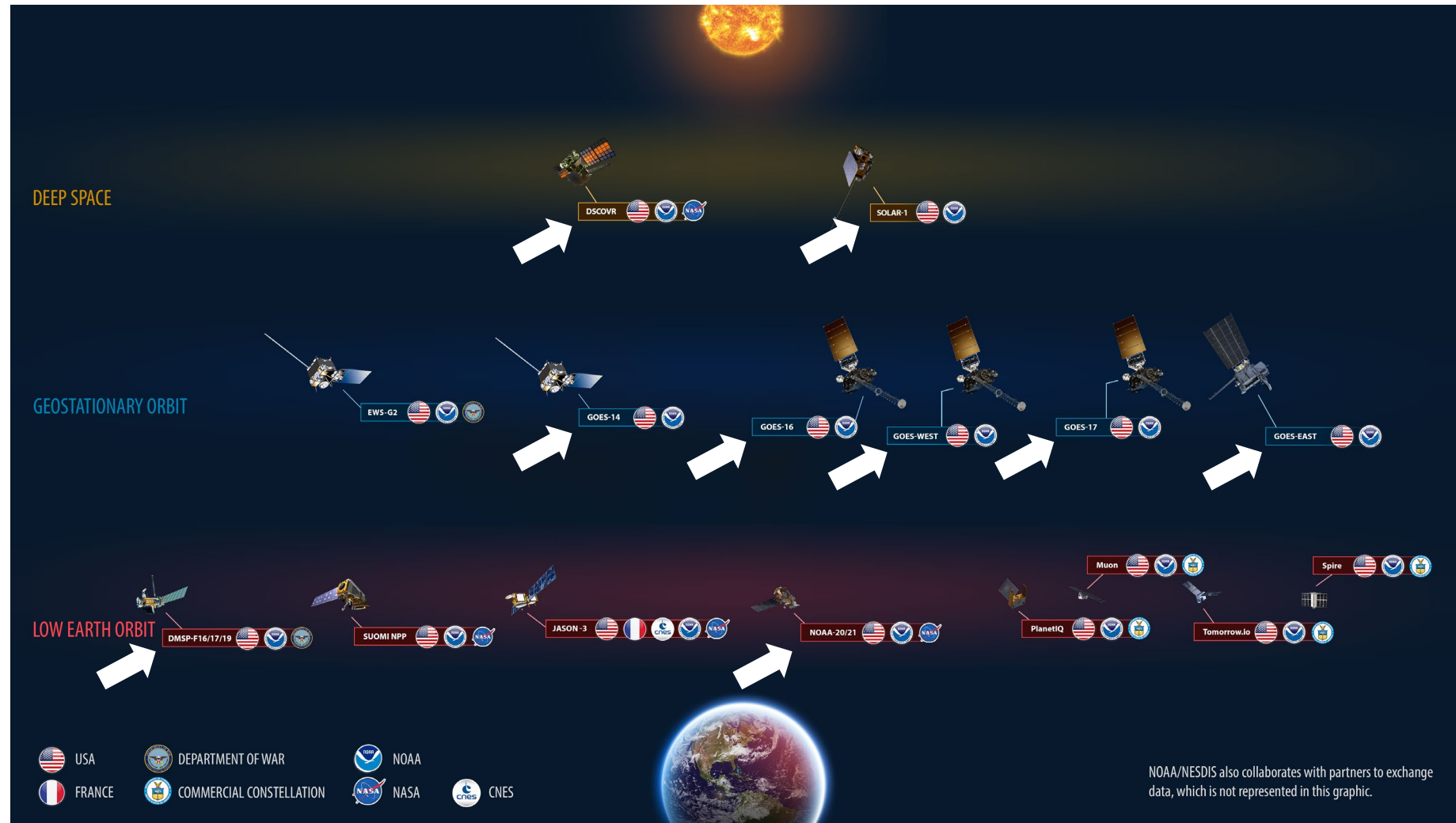
**Observations, Modeling,  
Research-to-Operations, and  
Decision Support**

**NESDIS National Centers  
for Environmental Information  
(NCEI)**



**Product Development and  
Quality, Archive & Access,  
Scientific Research**

# NOAA's Space Weather Satellites



# NOAA Satellite Data

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## LEO

- POES/Metop (since 1978)
- DMSP (1982-2025)

## MEO

- GPS/LANL (since 2000)

## GEO

- SMS/GOES (since 1974)

## Lagrange 1

- DSCOVR (since 2016)
- SOLAR (since 2025)

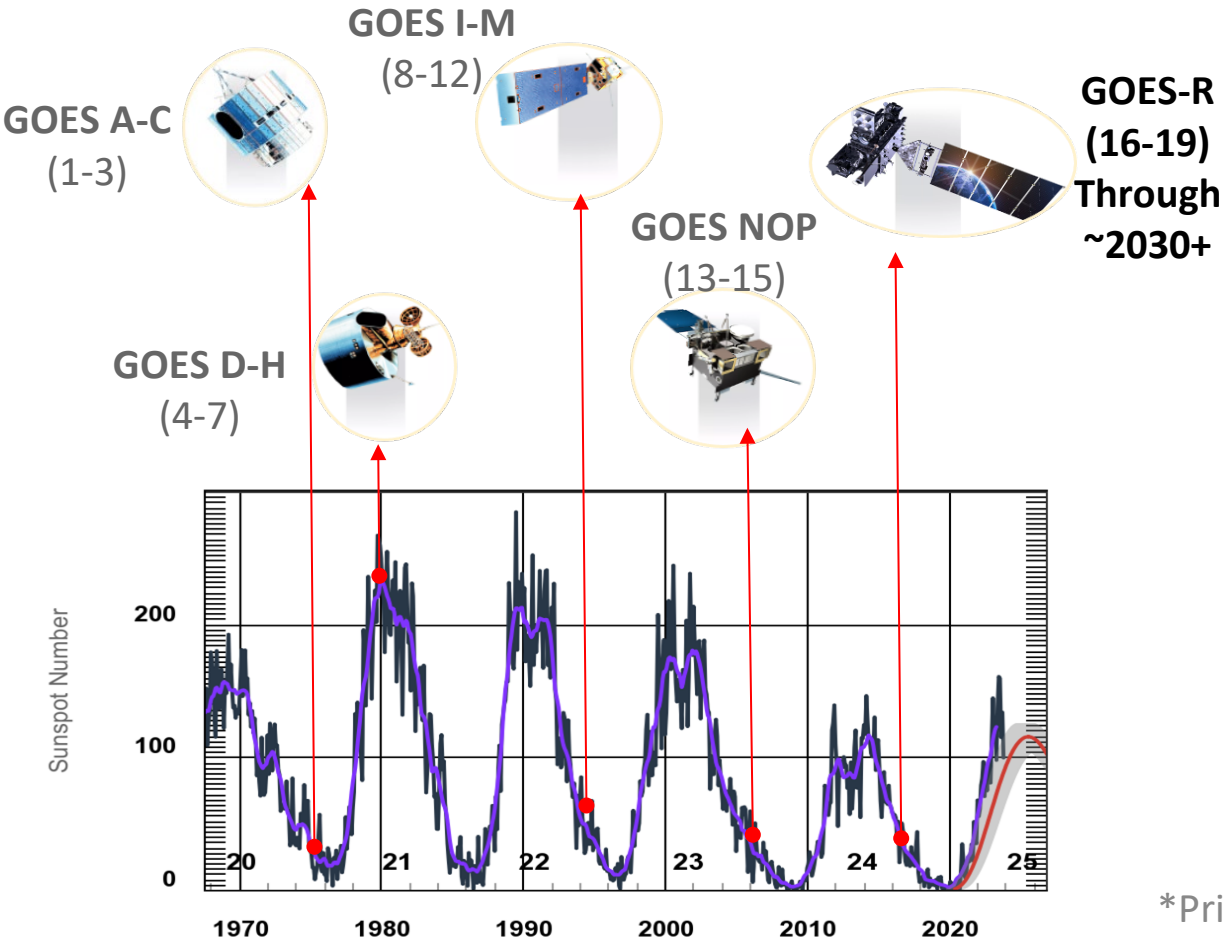
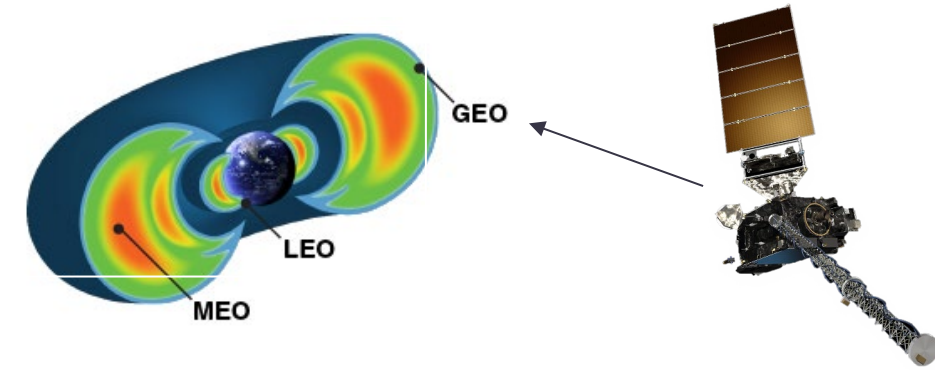
## Future- Space Weather Next Program

- Lagrange 1 – (SOLAR), launch ~2029\*
- Lagrange 5 – (Vigil, contributed coronagraph), launch ~2031\*
- GEO – Dedicated SWx satellite(s), planned\* for post GOES-R operations
- LEO – likely contributed instruments, in early preparative phases.

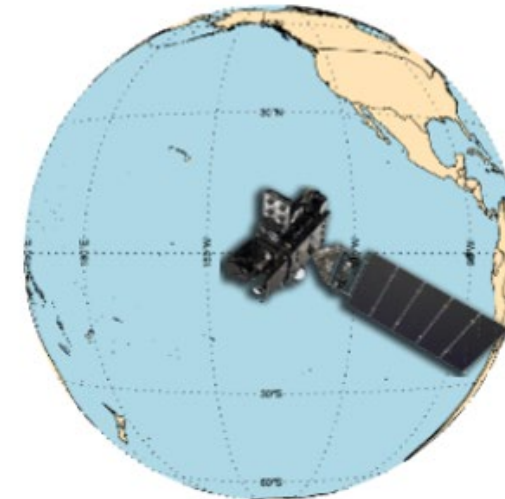
\* Notional dates, subject to change



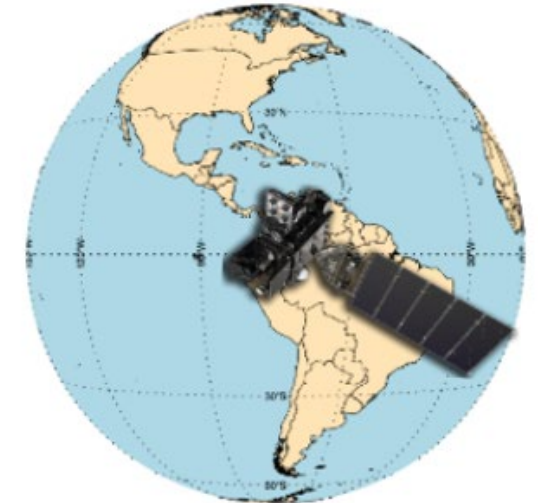
# GOES - Geostationary Operational Environmental Satellites



GOES-West  
GOES-18  
137.0°W



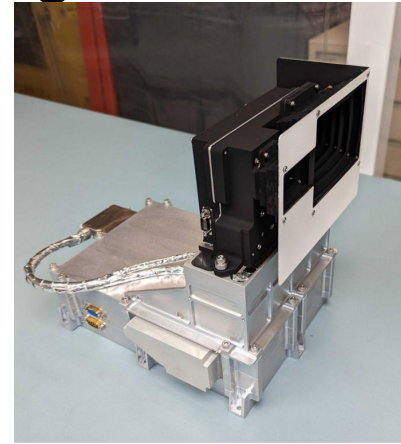
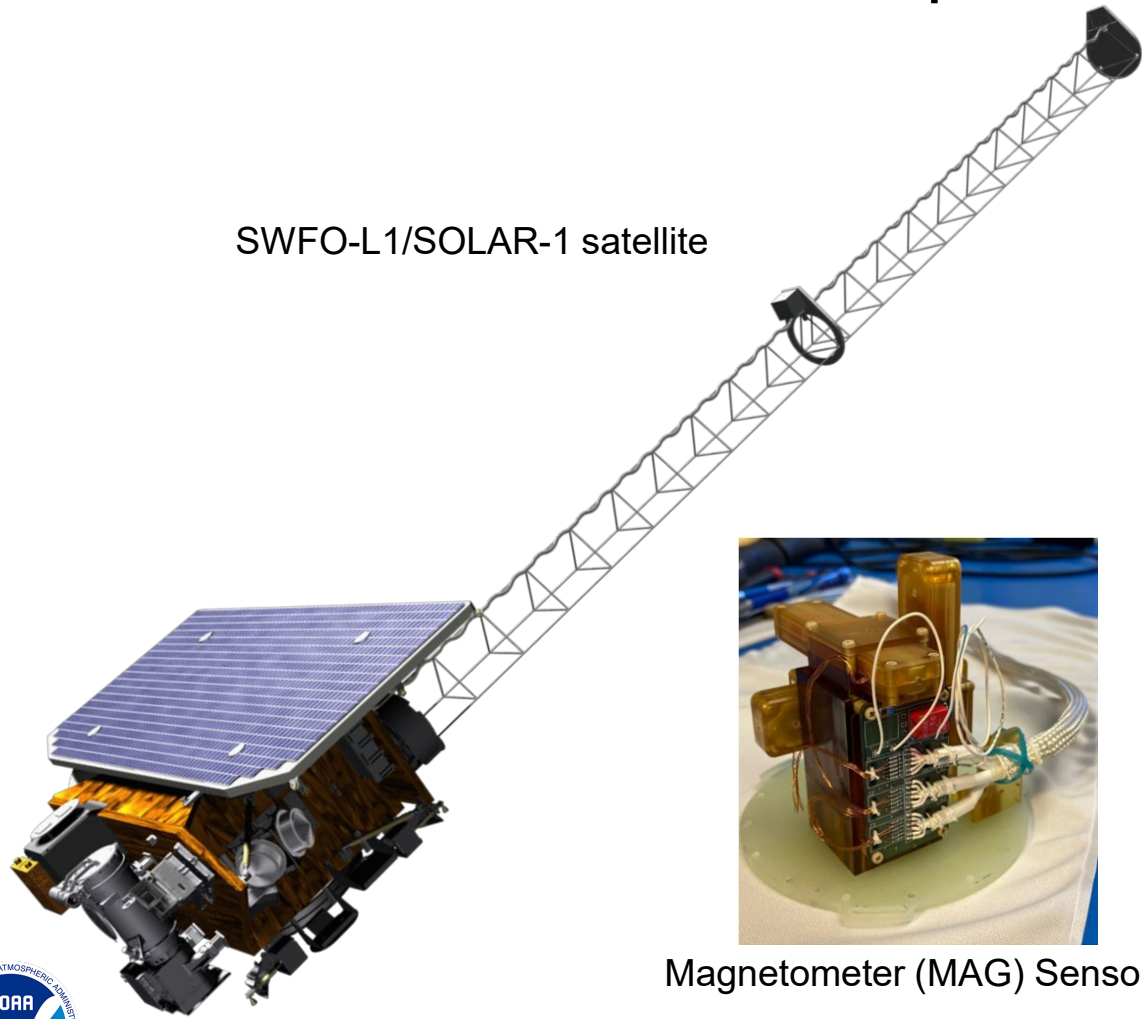
GOES-East  
GOES-16  
75.2°W



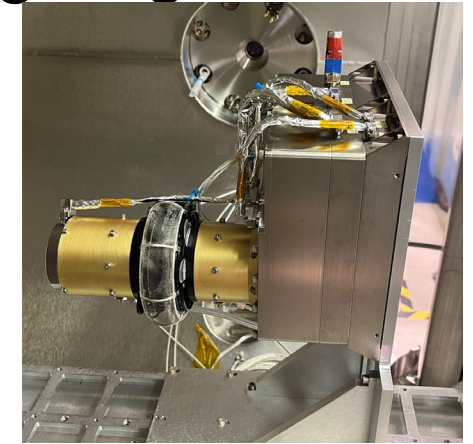
\*Prior to GOES (74-76): Synchronous Meteorological Satellite SMS-1 and SMS-2

# SWFO-L1, now SOLAR-1 (Space weather Observatory at L1 to Advance Readiness) launched Sept 2025. SOLAR-1 is the first of a satellite constellation to replace aging satellites at Lagrange 1

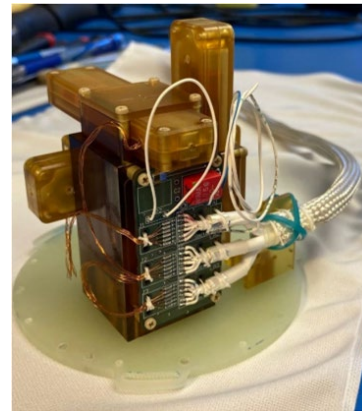
SWFO-L1/SOLAR-1 satellite



SupraThermal Ion Sensor (STIS)



Solar Wind Plasma Sensor (SWiPS)



Magnetometer (MAG) Sensor



Compact Coronagraph (CCOR-2) installation on SWFO-L1.

CCOR-1 is installed on GOES-19 as part of the SWFO Program.

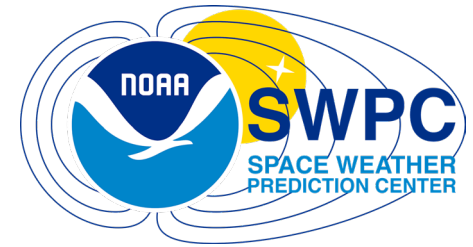


# NOAA's National Weather/Service Space Weather Prediction Center supports operational and real time users

Provides real-time and subscription access to NOAA space weather satellite data products, space weather model output, forecasts, warnings, and alerts.



<https://www.spaceweather.gov>





# NOAA's Space weather POrTal – **SPOT** – supports scientific users

Hosts SOLAR-1, DSCOVR, and SWPC model data. Other satellite and non-satellite space weather datasets are in the process of being incorporated.

<https://www.ncei.noaa.gov/cloud-access/space-weather-portal>

| Features                                        |                                |
|-------------------------------------------------|--------------------------------|
| Modern interface, easy navigation, mission info | Multiple data format download  |
| Data search                                     | 508-compliance, web styling    |
| Quicklook plots and movies                      | Product Monitoring reports     |
| Dynamic plotting tools                          | RESTful API (superset of HAPI) |



# NCEI Space Weather Products

## Legacy

Solar  
Photosphere

Legacy  
Publications

Solar  
Chromosphere

Cosmic Rays

Solar corona

Ionosphere

Solar flares  
and events

GeoMagnetic

Solar  
Irradiance

## Satellites

GOES 1-15

DMSP

GOES-R

DSCOVR

POES/MetOP

SWFO

## Partners

SWPC

AFRL

LANL

Hosting space weather data that spans 3 centuries!

<https://www.ncei.noaa.gov/products/space-weather>

recently reorganized for easier access

<https://doi.org/10.1029/2025SW004529>

